Bicycle Theft in the West Midlands

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Introduction:

While crime literature indicates a clear link between economic challenges and increased property crime, bicycle theft's relationship to the economy is a little more unclear (Raphael and Winter-Ebmer 2001, Christiansen 2026). Some authors model it as an "opportunity" crime (crime arising from increased opportunity/lowered risk of being penalized) as opposed to a "deprivation" crime (crime caused by economic hardship, Christiansen 2026). Having worked on analysis of the Dublin City Bikes, I knew weather was a pertinent factor impacting ridership, and sought to use it as a variable alongside unemployment to explore bicycle theft in the UK. The United Kingdom's West Midlands has extensive data on bicycle thefts reported to the county officials, which can be drawn in comparison with weather and unemployment statistics. Of course, the main underlying factor here is ridership/number of people utilizing bikes, which is not directly captured in the dataset (and presumably would be challenging to measure except via proxies or self-reported survey data).

The following report will explore, holding time and seasonal (monthly) effects constant, the relationship between meteorological factors, unemployment rates, and bicycle thefts in the West Midlands between 2013 and 2025. My research questions specifically are:

RQ 1: Do unemployment rates in the West Midlands area have a statistically significant relationship with bicycle theft rates, controlling for seasonal, meteorological, and time effects?

RQ 2: What methods are most appropriate for gauging the relationship of economic and meteorological factors with bicycle theft in the West Midlands?

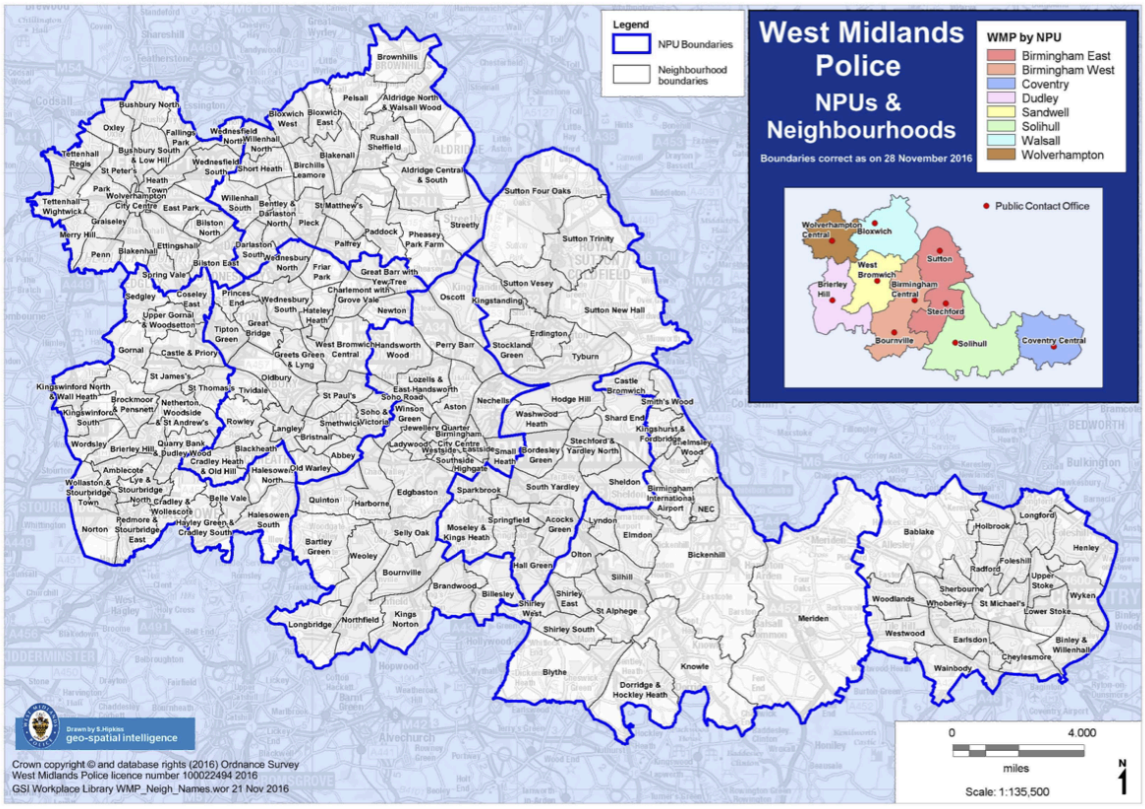
I will describe the datasets I chose for answering these questions and how I prepared them for analysis using Ordinary Least Squares and Random Forest regression, Principal Component Analysis, Lasso Regularization, and applied Double Machine Learning. I will discuss the outputs of these models and their implications for our understanding of bicycle theft

over the past decade in the West Midlands region, concluding with limitations of the study and recommendations for future inquiries.

Data Background:

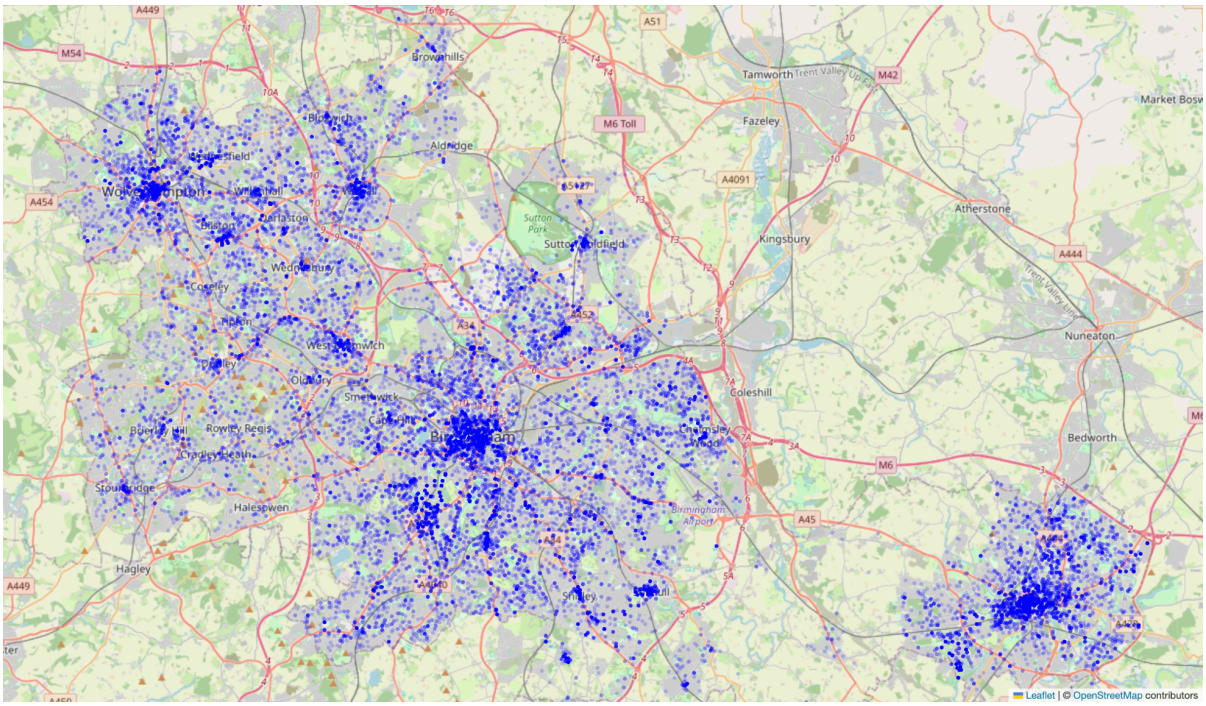
The UK Police (2026) collects crime data reported to all police jurisdictions across the UK. The West Midlands police jurisdiction, shown in Figure 1a, includes the city of Birmingham and surrounding suburban areas, along with a few townships. This means the dataset is representative of a mixture of rural-urban settings, though the mapped locations of data (see Figure 1b) show that the highest concentration of the crimes is reported in the central city areas.

Figure 1a: Map of West Midlands Police Jurisdictions



My weather data comes from the Shawbury meteorological station (Met Office 2025), and my unemployment data comes as a regional monthly average reported by the Office for National Statistics (2026). These datasets were chosen for their publication by official government or government affiliate sources, such that internal validity and missingness would be limited as much as possible.

Figure 1b: Map of bicycle thefts reported between June 2013 and December 2025.



Data Processing:

My data processing involved aligning all datasets to a common monthly timeframe, covering June 2013 to December 2025, where full overlap was available. Weather and unemployment variables were also transformed to include one-month lags, allowing for the analysis of potential delayed effects for each of my models. To control for seasonality, I one-hot encoded month indices. Treating months as numeric would impose an artificial ordering, while omitting them would leave substantial seasonal variation unaccounted for. Further, given the time-series nature of the data, a random train-test split was inappropriate, as it would ignore temporal ordering and autocorrelation among months. Instead, the final 20% of observations were reserved as a test set. The final dataset consists of monthly observations with the following variables: a time trend index, unemployment rate, air frost days, average maximum temperature, total sunlight hours, total rainfall, and total bicycle thefts, along with the monthly dummy variables.

Initial Data Visualizations:

Before anything else, I visualized the data to see what trends might be present. Figure 2a shows variables of interest over the timeframe of analysis. Zooming in on sunlight hours in comparison with bicycle theft in Figure 2b, we can see the seasonality lining up quite nicely.

Finally, we can see in Figure 2c that we might anticipate the highest bicycle theft rates to fall in October (month_10).

Figure 2a: Variables of interest over time:
X axis indicating months, beginning with June 2013 and ending with December 2025.

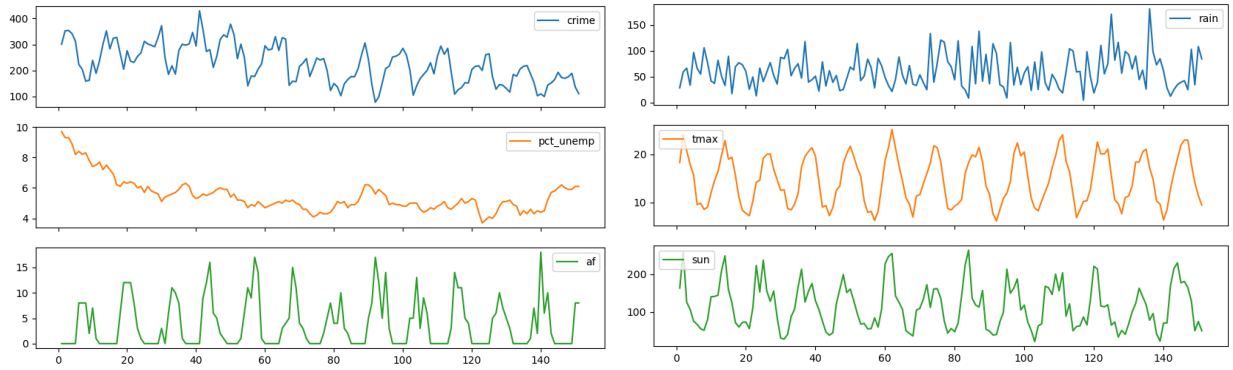


Figure 2b: Bicycle Theft and Sum Monthly Hours of Sunlight (logged on the right)

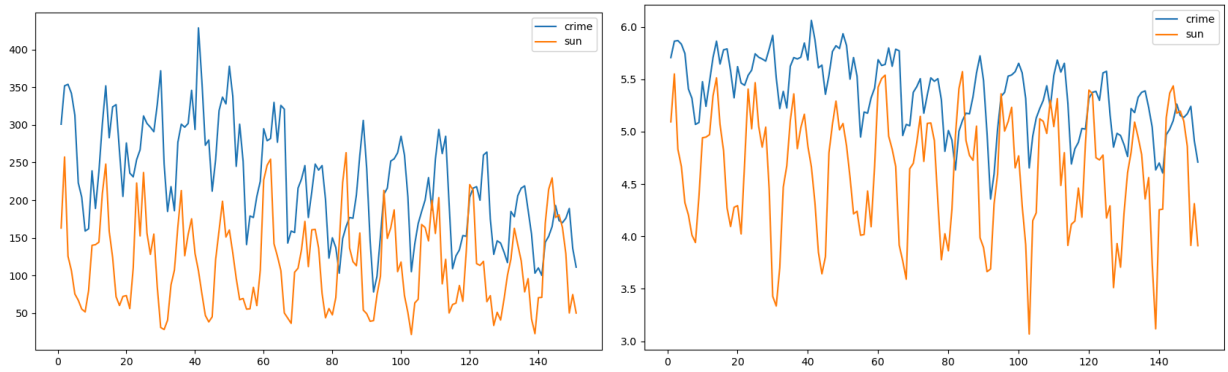
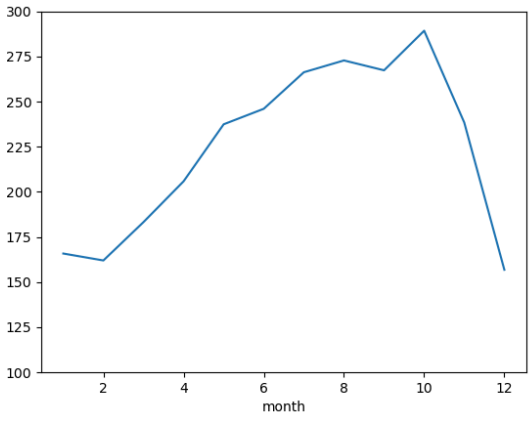


Figure 2c: Average Bicycle Theft Instances by Month



Methods and Justifications:

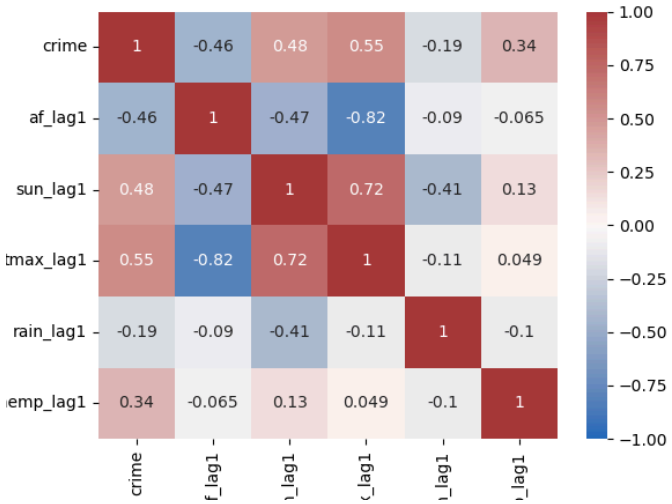
Lagged effects – For all of my methods, I trained and ran the models on non-lagged and again on once-lagged values of weather and unemployment variables to analyse their potential delayed effects on bicycle theft.

Ordinary Least Squares (OLS) – Utilizing OLS as the baseline estimation strategy for this project is appropriate because we would assume a linear relationship between input and outcome variables of interest (if any), and have controls in place for time trend and seasonality.

Random Forest – I also trained a Random Forest model to capture potential nonlinear relationships and interactions between weather variables and bicycle theft. This provides a robustness check against functional form misspecification in the OLS model, allowing assessment of whether relationships identified under linear assumptions persist under a flexible, nonparametric framework.

Principal Component Analysis (PCA) – I further ran PCA on the weather variables to mitigate their multicollinearity (see Figure 3) and identify which variables (at which weights) capture most of the variation in the weather dataset. This allows for more stable estimation in linear models, and serves as another opportunity to test for robustness of my OLS findings from the original feature set.

Figure 3: Lagged Variables Correlation Matrix



Lasso Regularization – After running PCA, I applied Lasso regularization to see which coefficients were penalized the most in my feature set. This allows for comparison between dimensionality reduction via transformation (PCA) and via selection (Lasso), and assessing whether similar variables are identified as most influential.

Double Machine Learning (DML): Finally, I employed cross-validated (k-fold) DML to estimate the partial effect of unemployment on bicycle theft while flexibly controlling for confounding variables. By using machine learning methods to residualize both the outcome and treatment variables, my DML models reduced the bias arising from nonlinear relationships. Five-fold cross-validation helped improve the stability and generalizability of the parameter estimates.

Results:

Across models, time trend, temperature, rainfall, and seasonal controls emerge as the primary predictors of bicycle theft. The time trend indicates that bicycle theft has declined over the sample period, while monthly fixed effects reveal clear seasonal patterns, with theft increasing during summer months relative to the reference category (January). Among weather variables, rainfall is consistently statistically significant in OLS specifications, indicating a strong negative association with bicycle theft. This aligns with the intuition that adverse weather conditions reduce bicycle usage and therefore opportunities for theft. In contrast, unemployment does not exhibit statistically significant effects in the OLS framework, although it approaches significance at the 10% level in some specifications. Notably, however, Lasso regularization does not shrink the unemployment coefficient to zero, suggesting that it on its own is not very strong, but perhaps alternative economic metrics could have some relevance.

Table 1: OLS Results: Monthly Predictors of Bicycle Theft
(all lagged once, with monthly fixed effects)

Coefficients		Standard Error	P-Value	Addt. Model Stats
Constant	276.32	62.21	0.00	R^2: 0.72
Time Trend	-1.16	0.17	0.00	RMSE: 24.08
Unemployment	-7.97	4.68	0.10	F-stat: 15.28
Rain (total)	-0.35	0.15	0.03	F-p-value: 6.7 e^{-21}
Max Temp (℃)	3.42	4.30	0.43	AIC: 1249
Sunlight Hours	-0.18	0.17	0.28	BIC: 1300
Frost	1.73	1.93	0.37	N: 120

Note: variables significant beyond the $p<0.05$ level are highlighted

PCA revealed that the significance of weather variables is sensitive to how seasonality is modeled. In my models, components with heavy weighting on rainfall (0.87 on PC2 (no lag) and

PC4 (lag)) were often statistically significant in the OLS models (see Table # 2), while temperature-heavy components (0.62 on PC1 (no lag) and PC3 (lag)) didn't show significance. More specifically:

- In models with monthly controls and evaluated on test data, rainfall-dominated components retained significance.
- In models using the full dataset without monthly controls, both temperature- and rainfall-heavy components appeared significant.

Table 2: OLS Results with PCA on Weather Predictors
(all lagged once, with monthly fixed effects)

Coefficients		Standard Error	P-Value	Addt. Model Metrics
Constant	276.41	40.13	0.00	R^2: 0.71
Time Trend	-1.09	0.16	0.00	RMSE: 23.35
Unemployment	-6.82	4.88	0.17	F-stat: 17.04
PC # 3 (tmax heavy)	-1.74	7.42	0.83	F-p-value: 1.3 e^{-21}
PC # 4 (rain heavy)	-9.60	4.19	0.02	AIC: 1248

Note: variables significant beyond the $p<0.05$ level are highlighted

When applying Double Machine Learning to isolate the independent effects of weather variables, a different pattern emerges. After flexibly controlling for confounding factors, temperature becomes the dominant predictor, with substantially higher explanatory power than rainfall. When isolating each variable individually, average maximum temperature performs approximately twice as well as rainfall in terms of R^2 values, and the PC's with temperature-heavy weighting likewise perform better than their rain-heavy counterparts.

Table 3: Double Machine Learning Model Result Comparisons
Models color-coded for their increasing R^2 values (green>yellow>red)

Model Name	Predictor(s)	Controls	R^2
dml	Unemployment	Weather, time trend, monthly seasonality	0.001
dml_lag	Unemployment (lagged once)	Weather (lagged), time trend, monthly seasonality	0.012
dml_pca	Unemployment	PC #1, PC #2, time trend monthly seasonality	0.000
dml_pca_lag	Unemployment (lagged once)	PC #3, PC #4, time trend, monthly seasonality	0.004

dml_weather	PC #1 (tmax-heavy, no lag)	Unemployment, time trend, monthly seasonality	0.274
dml_weather_lag	PC #3 (tmax-heavy, lag)	Unemployment (lagged), time trend, monthly seasonality	0.268
dml_tmax	Avg Maximum Temperature	Weather, unemployment, time trend, monthly seasonality	0.114
dml_tmax_lag	Avg Maximum Temperature	Weather (lagged), unemployment (lagged), time trend, monthly seasonality	0.034

Table 4: Double Machine Learning Results for PC#3, with Lag			
Coefficients		Standard Error	P-Value
Constant	-1.19	4.37	0.79
PC #3 (tmax-heavy)	25.99	3.96	0.00
Additional Model Metrics:			
F-stat: 43.14		AIC: 1271	R^2: 0.27
F-p-value: 1.4 e^{-09}		BIC: 1277	N: 120
Note: variables significant beyond the $p<0.05$ level are highlighted			

Discussion:

In the OLS framework, rainfall emerges as the most consistently significant weather variable. However, this likely reflects its role as a proxy for broader seasonal and environmental conditions. Rainfall is strongly correlated with other aspects of weather and time of year, making it an effective summary measure of "unfavorable cycling conditions." By contrast, the DML framework shows temperature emerging as the more important predictor, suggesting that it plays a more direct behavioral role. Higher temperatures likely increase bicycle usage, exposure, and time spent outdoors, thereby increasing opportunities for theft. Rainfall, on the other hand, appears to operate more indirectly by reflecting general weather conditions rather than exerting a distinct independent effect.

The lack of strong significance for unemployment across models suggests that economic conditions play a more limited or indirect role in explaining bicycle theft in the West Midlands. While Lasso results indicate that unemployment is not entirely irrelevant (the unemployment coefficient was not zeroed), its effect appears weaker and less robust than that of environmental factors. This may reflect the nature of bicycle theft as an opportunity-driven crime, more strongly influenced by exposure and routine activity patterns than by broader economic conditions.

Alternatively, it may indicate that unemployment is an imperfect proxy for economic stress, and that other measures (e.g., income, deprivation indices) could yield different results.

More broadly, the comparison across methods demonstrates the importance of model choice in empirical analysis. Linear models such as OLS are valuable for interpretation but can be sensitive to multicollinearity and functional form assumptions. Machine learning approaches, including Random Forests and DML, provide more flexible modeling of complex relationships and can reveal patterns that are not apparent under linear specifications. The divergence between OLS and DML results in this study illustrates how different methods can uncover different aspects of the data-generating process.

Limitations:

This project has many limitations. First, reporting of bicycle thefts introduces a selection bias against those who do not choose to report to police. Second, because of my inability to control for ridership levels fluctuations, I cannot claim direct causality–merely correlation with controls for confounding variables. Third, most of my variables seem most powerful as proxies for broader concepts (unemployment as a proxy for economic conditions, rain as a proxy for poor/cold weather). Many other proxies could be used to represent the same phenomena, and perhaps represent them better. Finally, I lost granularity in my aggregation to monthly data, and the vastness of the West Midlands region doesn't permit a clear delineation between urban and suburban/rural trends.

Policy Implications, Recommendations:

My project shows that utilizing unemployment as a proxy for identifying the link between economic conditions and bicycle theft is unlikely to yield actionable results. It also supports the growing corpus of literature suggesting that bicycle theft is much more an "opportunity" crime than a "deprivation" one. I would recommend future research focus on seasonal and spatial patterns (akin to Christiansen 2026). For those seeking to predict bicycle theft, I would recommend the use of a time-series specific model such as ARIMA.

Besides future policy research, my project can only indicate that West Midlands governments could coordinate their expenditures on anti-bicycle-theft informational advertisements with seasonal changes (hotter months would be best for higher expenditures).

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All data sources are publicly available under open data licenses.

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